

JEE MAIN REVISION MASTER: ELECTROMAGNETIC INDUCTION 2026 2 QUESTIONS 3.33% WEIGHTAGE $\uparrow$ 99.4%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 21 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Electromagnetic Induction - Core Application	21 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Electromagnetic Induction - Core Application (Very High Yield — 21 PYQs)

Core Principle: Key problem pattern in Kcet Electromagnetic Induction.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In the figure shown, the conductor PQ of length l is moved from $x=0$ to $x=b$ and then up to $x=2b$ with a constant velocity $\vec{v}$ " [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "In Faraday-Henry's experiment, a coil is connected to a galvanometer. For the deflection of pointer in the galvanometer, which of the follow..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "When a bar magnet is pushed towards the coil, along its axis, as shown in the figure, the galvanometer pointer deflects towards X . When thi..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In the figure shown, the conductor PQ of length l is moved from $x=0$ to $x=b$ and then up to $x=2b$ with a constant velocity $\vec{v}$. A uniform magnetic field $\vec{B}$ is perpendicular to the plane of the paper and extends from $x=0$ to $x=b$ and it is zero from $x>b$. The magnitude of emf Induced in the conductor is

- (A) $Blv; 0 \leq x \leq b$
- (B) Zero; $0 \leq x \leq b$
- (C) $Blv; 0 \leq x \leq b$
- (D) $Blv; b \leq x \leq 2b$

Correct Answer: None

Solution Analysis: When conductor PQ moves from $x=0$ to $x=b$, it is inside the uniform magnetic field $\vec{B}$. The magnetic field is perpendicular to the plane of the paper. The conductor has length l and moves with constant velocity $\vec{v}$. For a straight conductor moving perpendicular to a magnetic field, the motional emf induced across its ends is $E=Blv$. Here, B , l , and v are constant. So, the induced emf remains constant while the conductor is in the region $0 \leq x \leq b$. When the conductor moves from $x=b$ to $x=2b$, it is outside the magnetic field region. In this region, the magne